

# Exact inverse-hypergeometric coefficient certificates through degree 39 and a repaired 1.0085 Krivine constant

Candidate partial result for arXiv:1804.03644  
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**Status: exact computer-assisted partial result, likely valid; human verification requested.** The inverse-coefficient conditions conjectured in the source are certified for every odd degree through 39. Combined with the source’s analytic tail lemma, this repairs a false final numerical implication in the published argument and proves the slightly stronger factor 1.0085. The all-orders endpoint-extremality conjecture remains open.

## 1 Source question and result

For  $a, b \in [0, 1]$ , put

$$\alpha = \frac{1-a}{2}, \quad \beta = \frac{1-b}{2}, \quad F_{a,b}(z) = z {}_2F_1\left(\alpha, \beta; \frac{3}{2}; z^2\right).$$

Write the local compositional inverse at the origin as

$$G_{a,b}(w) = F_{a,b}^{-1}(w) = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} d_k(a,b) w^k, \quad h_{a,b}(r) = \sum_{\substack{k \geq 1 \\ k \text{ odd}}} |d_k(a,b)| r^k.$$

Bhattiprolu, Ghosh, Guruswami, Lee, and Tulsiani conjectured that the Grothendieck endpoint is extremal [1]:

$$h_{a,b}^{-1}(1) \geq \operatorname{arsinh}(1) \quad (a, b \in [0, 1]). \quad (1.1)$$

Their statement is reproduced below.

**Transcription of the question.** Is  $h_{a,b}^{-1}(1) \geq \operatorname{arsinh}(1)$  for every  $a, b \in [0, 1]$ ?

The source observed that the following two coefficient conditions, if true in every odd degree, imply (1.1):

$$d_k(a,b) \leq \frac{1}{k!} \quad \text{if } k \equiv 1 \pmod{4}, \quad (C1)$$

$$d_k(a,b) \leq 0 \quad \text{if } k \equiv 3 \pmod{4}. \quad (C2)$$

It reports an exact Mathematica verification through  $k = 29$  and then invokes a uniform tail bound. We extend the exact range by ten degrees and obtain a rigorous quantitative repair.

**Theorem 1** (Degree-39 certificate and repaired constant). *For every  $a, b \in [0, 1]$ , conditions (C1) and (C2) hold for every odd  $k \leq 39$ . Consequently,*

$$h_{a,b}^{-1}(1) \geq 0.874 \geq \frac{\operatorname{arsinh}(1)}{1.0085}. \quad (1.2)$$

*Every computation in the coefficient certificate and in the numerical comparison is over the rational numbers.*

decreasing in  $a$  and in  $b$ .

(M4) Note that  $f_{a,b}(0) = 0$  and by (M1) and (M2),  $f_{a,b}(1) \geq 1$ . By continuity,  $f_{a,b}([0, 1])$  contains  $[0, 1]$ . Combining this with (M3) implies that for any fixed  $\rho \in [0, 1]$ ,  $f_{a,b}^{-1}(\rho)$  is increasing in  $a$  and in  $b$ .

#### 4 $\sinh^{-1}(1)/(1 + \varepsilon_0)$ Bound on $h_{a,b}^{-1}(1)$

In this section we show that  $p = \infty, q = 1$  (the Grothendieck case) is roughly the extremal case for the value of  $h_{a,b}^{-1}(1)$ , i.e., we show that for any  $1 \leq q \leq 2 \leq p \leq \infty$ ,  $h_{a,b}^{-1}(1) \geq \sinh^{-1}(1)/(1 + \varepsilon_0)$  (recall that  $h_{0,0}^{-1}(1) = \sinh^{-1}(1)$ ). While we were unable to establish as much, we conjecture that  $h_{a,b}^{-1}(1) \geq \sinh^{-1}(1)$ . Section 4.1 details some of the challenges involved in establishing that  $\sinh^{-1}(1)$  is the worst case, and presents our approach to establish an approximate bound, which will be formally proved in Section 4.2.

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Figure 1: Full-width crop of physical PDF page 20 (paper page 18) of arXiv:1804.03644. The authors state that they could not establish the endpoint bound and conjecture (1.1).

## 2 Proof intuition

Lagrange inversion turns each  $d_{2n+1}$  into one coefficient of a negative integer power of a hypergeometric series. After substituting  $\alpha = (1 - a)/2$  and  $\beta = (1 - b)/2$ , this is an exact rational polynomial on the square  $[0, 1]^2$ . Positivity on that square can be certified without sampling: convert the target polynomial to the tensor-product Bernstein basis. If every Bernstein coefficient is nonnegative, the polynomial is nonnegative everywhere. This succeeds in every required degree through 39 except  $k = 7$ , for which a short two-variable inequality gives a stronger strict certificate.

The source’s defect lemma then allows all unknown coefficients from degree 41 onward to be paid for by one geometric tail. Extending the exact range from 29 to 39 makes that tail small enough to prove  $h^{-1}(1) \geq 0.874$ . Finally, short rational Taylor bounds for  $\sinh$  certify that 0.874 is at least  $\operatorname{arsinh}(1)/1.0085$ . No floating-point comparison enters the proof.

## 3 Exact coefficient certificate

Set

$$\Phi(t) = {}_2F_1\left(\alpha, \beta; \frac{3}{2}; t\right) = \sum_{j \geq 0} c_j t^j, \quad c_j = \frac{(\alpha)_j (\beta)_j}{(3/2)_j j!}.$$

Lagrange inversion gives, for  $n \geq 0$ ,

$$d_{2n+1} = \frac{1}{2n+1} [t^n] \Phi(t)^{-(2n+1)}. \quad (3.1)$$

Thus every coefficient needed in Theorem 1 is an exactly computable element of  $\mathbb{Q}[a, b]$ .

We recall the elementary positivity certificate used by the verifier. If

$$P(a, b) = \sum_{u=0}^r \sum_{v=0}^s p_{uv} a^u b^v,$$

then its coefficients in the bidegree- $(r, s)$  Bernstein basis are

$$B_{ij} = \sum_{u \leq i} \sum_{v \leq j} p_{uv} \frac{\binom{i}{u}}{\binom{r}{u}} \frac{\binom{j}{v}}{\binom{s}{v}}. \quad (3.2)$$

The identity

$$P(a, b) = \sum_{i=0}^r \sum_{j=0}^s B_{ij} \binom{r}{i} a^i (1-a)^{r-i} \binom{s}{j} b^j (1-b)^{s-j}$$

shows that  $B_{ij} \geq 0$  for every  $i, j$  implies  $P \geq 0$  on  $[0, 1]^2$ .

For  $k \equiv 1 \pmod{4}$ , the exact verifier applies (3.2) to

$$\frac{1}{k!} - d_k(a, b). \quad (3.3)$$

For  $k \equiv 3 \pmod{4}$ , it first cancels the manifest nonnegative edge factor and applies (3.2) to

$$\frac{-d_k(a, b)}{(1-a)(1-b)}. \quad (3.4)$$

Using (3.1), every Bernstein coefficient in (3.3)–(3.4) is computed as an exact rational number. They are all nonnegative for

$$k = 3, 5, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39.$$

The case  $k = 1$  is the identity  $d_1 = 1$ . Only  $k = 7$  is not certified by the direct Bernstein basis (two coefficients are negative), and it is settled next.

**Lemma 2** (The exceptional seventh coefficient). *For every  $a, b \in [0, 1]$ , one has  $d_7(a, b) < 0$  unless  $a = 1$  or  $b = 1$ , in which case  $d_7 = 0$ .*

*Proof.* Exact expansion of (3.1) and factorization give

$$-d_7 = \frac{(1-a)(1-b)}{5040} P(a, b),$$

where, on writing  $u = a + b$  and  $v = ab$ ,

$$P = 225v^2 - 344uv + 127u^2 + 34v - 8u + 1. \quad (3.5)$$

Suppose first that  $0 \leq u \leq 11/100$ . Since  $0 \leq v \leq u^2/4$ ,

$$P \geq 1 - 8u - 86u^3 \geq 1 - \frac{88}{100} - 86 \left( \frac{11}{100} \right)^3 > 0. \quad (3.6)$$

Now suppose  $11/100 \leq u \leq 2$ . On the feasible interval for  $v$ ,

$$\frac{\partial P}{\partial v} = 450v - 344u + 34 \leq \frac{225}{2}u^2 - 344u + 34 < 0. \quad (3.7)$$

Indeed, the last quadratic is convex and is negative at both endpoints  $11/100$  and  $2$ , hence is negative throughout the interval. Therefore (3.5) is minimized at the largest possible  $v$ , namely  $v = u^2/4$ ; equivalently  $a = b = t = u/2$ . At that point

$$\begin{aligned} P(t, t) &= 225t^4 - 688t^3 + 542t^2 - 16t + 1 \\ &= \left( 15t^2 - \frac{344}{15}t + \frac{15}{43} \right)^2 + \frac{54152}{9675}t^2 + \frac{1624}{1849} > 0. \end{aligned} \quad (3.8)$$

Equations (3.5)–(3.8) prove the claim.  $\square$

This completes the exact proof of (C1)–(C2) through degree 39.

## 4 Tail estimate and the constant

The source proves the inverse-coefficient estimate

$$|d_k(a, b)| \leq \frac{6.1831}{k} \quad (4.1)$$

uniformly for the present purpose; its displayed result is stronger in the interior of the parameter square [1, Lemma 4.7]. Take

$$\delta = \frac{439}{500} = 0.878, \quad E = \frac{61831}{10000 \cdot 41} \frac{\delta^{41}}{1 - \delta^2}. \quad (4.2)$$

Since only odd powers occur, (4.1) gives

$$h_{\text{err}}(41, \delta) := \sum_{\substack{k \geq 41 \\ k \text{ odd}}} |d_k| \delta^k \leq E. \quad (4.3)$$

Put  $\rho = \text{arsinh}(1 - 2E)$ . The exact inequalities below show in particular that  $\rho \leq \delta$ . The proof of the source's defect lemma remains valid when an upper bound  $E \geq h_{\text{err}}(41, \delta)$  replaces the exact tail: using (C1)–(C2) below degree 41,

$$\begin{aligned} h_{a,b}(\rho) &\leq -G_{a,b}(\rho) + \sin \rho + \sinh \rho + 2h_{\text{err}}(41, \rho) \\ &\leq -G_{a,b}(\rho) + \sin \rho + 1 + 2(h_{\text{err}}(41, \rho) - E) \\ &\leq -G_{a,b}(\rho) + \sin \rho + 1 \leq 1. \end{aligned} \quad (4.4)$$

The last inequality is precisely the monotonicity step in the source's Lemma 4.1:  $G_{a,b}(\rho) \geq G_{0,0}(\rho) = \sin \rho$ . Hence  $h_{a,b}^{-1}(1) \geq \rho$ .

It remains only to compare three rational quantities. For rational  $0 \leq x \leq 1$ , define

$$S_N(x) = \sum_{j=0}^N \frac{x^{2j+1}}{(2j+1)!}.$$

Positivity of the Taylor series and a geometric majorant for successive terms give

$$S_N(x) \leq \sinh x \leq S_N(x) + \frac{x^{2N+3}}{(2N+3)!} \left( 1 - \frac{x^2}{(2N+4)(2N+5)} \right)^{-1}. \quad (4.5)$$

With  $N = 12$ , exact rational arithmetic in (4.2) and (4.5) verifies

$$\sinh\left(\frac{437}{500}\right) \leq 1 - 2E \leq \sinh\left(\frac{439}{500}\right), \quad (4.6)$$

and

$$\sinh\left(\frac{437}{500} \frac{2017}{2000}\right) \geq 1. \quad (4.7)$$

Thus  $0.874 \leq \rho \leq 0.878$  and  $\text{arsinh}(1) \leq 0.874 \cdot 1.0085$ . Combining this with (4.4) proves Theorem 1.

## 5 The repaired arithmetic step

The source’s Theorem 4.2 advertises  $\varepsilon_0 = 0.00863$ , but its final displayed step is

$$\operatorname{arsinh}(0.974202) \geq \frac{\operatorname{arsinh}(1)}{1 + 0.00863}. \quad (5.1)$$

This implication is false: the two sides are approximately 0.863013 and 0.873832, respectively. The relevant source page is reproduced below.

$$\begin{aligned} &= -f_{0,0}^{-1}(\rho) + \sin(\rho) + 1 && \text{(Corollary 3.19 : (M4))} \\ &= 1 && (f_{0,0}^{-1}(\rho) = \sin(\rho)) \quad \blacksquare \end{aligned}$$

Thus we obtain,

**Theorem 4.2.** *For any  $1 \leq q \leq 2 \leq p \leq \infty$ , let  $a := p^* - 1, b = q - 1$ . Then for any  $m, n \in \mathbb{N}$  and  $A \in \mathbb{R}^{m \times n}$ ,  $\operatorname{CP}(A)/\|A\|_{p \rightarrow q} \leq 1/(h_{a,b}^{-1}(1) \cdot \gamma_q \gamma_{p^*})$  and moreover*

$$\begin{aligned} &- h_{1,b}^{-1}(1) = h_{a,1}^{-1}(1) = 1. \\ &- h_{a,b}^{-1}(1) \geq \sinh^{-1}(1)/(1 + \varepsilon_0) \text{ where } \varepsilon_0 = 0.00863. \end{aligned}$$

*Proof.* The first inequality follows from Lemma 2.7. As for the next item, If  $p = 2$  or  $q = 2$  (i.e.,  $a = 1$  or  $b = 1$ ) we are trivially done since  $h_{a,b}^{-1}(\rho) = \rho$  in that case (since for  $k \geq 1$ ,  $(0)_k = 0$ ). So we may assume that  $a, b \in [0, 1)$ .

We are left with proving the final part of the claim. Now using Mathematica we verify (exactly)<sup>2</sup> that (C1) and (C2) are true for  $k \leq 29$ . Now let  $\delta = \sinh^{-1}(0.974203)$ . Then by Lemma 4.7 (which states that  $f_k^{-1}$  decays exponentially and will be proven in the subsequent section),

$$h_{err}(31, \delta) := \sum_{k \geq 31} |f_k^{-1}| \cdot d^k \leq \frac{6.1831}{31} \cdot \frac{\delta^{31}}{1 - \delta^2} \leq 0.0128991 \dots$$

Now by Lemma 4.1 we know  $h_{a,b}^{-1}(1) \geq \sinh^{-1}(1 - 2h_{err}(31, \delta))$ . Thus,

$$h_{a,b}^{-1}(1) \geq \sinh^{-1}(0.974202) \geq \sinh^{-1}(1)/(1 + \varepsilon_0) \text{ for } \varepsilon_0 = 0.00863, \text{ which completes the proof. } \blacksquare$$

### 4.2 Bounding Inverse Coefficients

In this section we prove that  $f_k^{-1}$  decays as  $1/c^k$  for some  $c = c(a, b) > 1$ , proving Lemma 4.7. Throughout this section we assume  $1 \leq p^*, q < 2$ , and  $a = p^* - 1, b = q - 1$  (i.e.,  $a, b \in [0, 1)$ ). Via

<sup>2</sup> We generated  $f_k^{-1}$  as a polynomial in  $a$  and  $b$  and maximized it over  $a, b \in [0, 1]$  using the Mathematica “Maximize” function which is exact for polynomials.

Figure 2: Full-width crop of physical PDF page 22 (paper page 20) of arXiv:1804.03644, including Theorem 4.2, the degree-29 computation, the tail bound, and the final numerical implication (5.1).

Theorem 1 is not a reinterpretation of (5.1): the additional exact coefficients reduce the tail at degree 41, and (4.5)–(4.7) replace the false floating-point comparison by rational inequalities. It therefore repairs the advertised theorem and slightly strengthens its approximation factor from 1.00863 to 1.0085.

## 6 Verification, limitations, and novelty

- The accompanying verifier independently constructs every  $d_k$  from (3.1), rather than importing a table. It performs the substitutions, factor cancellations, Bernstein conversion, and sign tests

in exact SymPy rational arithmetic. The  $k = 7$  factorization and every constant comparison in (4.6)–(4.7) are also asserted exactly.

- The certificate is finite and computer-assisted. It proves no sign condition for  $k \geq 41$ , and it does not settle the all-orders conjecture (1.1). The analytic tail estimate and the monotonicity step in (4.4) are taken from the source.
- A numerical scan through inverse degree 121 on a  $21 \times 21$  parameter grid found no counterexample to (1.1). A stronger experimental coefficientwise inequality  $G_{a,b}^{(4)} \leq G_{a,b}$  also held on that grid, but neither observation is used in the theorem.
- A bounded novelty search on 22 July 2026 used arXiv/web searches for the exact title, “generalized Krivine rounding”, “ $h_{a,b}^{-1}(1)$ ”, and the inverse-hypergeometric coefficient conditions. It found the source paper but no later proof or counterexample to (1.1), nor an erratum repairing Theorem 4.2. This is evidence, not a definitive literature certification.

**Human-review recommendation.** Verify and retain as a quantitative partial result. The highest-value checks are the exact execution of the verifier, the  $k = 7$  reduction (3.5)–(3.8), and the use of the source’s Lemmas 4.1 and 4.7 in (4.1)–(4.4).

## References

- [1] V. Bhattiprolu, M. Ghosh, V. Guruswami, E. Lee, and M. Tulsiani, *Approximating operator norms via generalized Krivine rounding*, arXiv:1804.03644 (2018).